

Desktop.Space

Intelligent desktop context awareness and automation with eye tracking, messaging pipeline, and Moire vision.

Overview

Desktop.Space gives Vibemind eyes and hands on the user's desktop. The space operates across multiple integrated systems: eye/gaze tracking via webcam (MediaPipe), screen analysis, automated mouse/keyboard control, and a messaging pipeline for chat automation. The eyeterm subsystem alone encompasses 15 subdirectories.

Backend Agent: DesktopAgent (21 Events)

File: `python/spaces/desktop/agents/desktop_agent.py`

Desktop Operations (7 Events)

Event	Tool Function	Description
<code>desktop.open_app</code>	<code>open_app</code>	Open application
<code>desktop.click</code>	<code>click_element</code>	Click element
<code>desktop.type</code>	<code>type_text</code>	Type text
<code>desktop.press_key</code>	<code>press_key</code>	Press key
<code>desktop.screenshot</code>	<code>take_screenshot</code>	Take screenshot
<code>desktop.scroll</code>	<code>scroll_screen</code>	Scroll
<code>desktop.task</code>	<code>execute_desktop_task</code>	Execute desktop task

Task Management (3 Events)

Event	Tool Function
<code>desktop.task.create</code>	<code>create_task_node</code>
<code>desktop.task.update</code>	<code>update_task_status</code>
<code>desktop.task.list</code>	<code>get_task_list</code>

Moire Vision (2 Events)

Event	Tool Function
<code>desktop.moire.scan</code>	<code>moire_scan</code>
<code>desktop.moire.find</code>	<code>moire_find_element</code>

Messaging / Clawdbot Bridge (7 Events)

Event	Tool Function
<code>messaging.whatsapp</code>	<code>send_whatsapp</code>
<code>messaging.telegram</code>	<code>send_telegram</code>
<code>messaging.send</code>	<code>send_message</code>
<code>web.search</code>	<code>web_search</code>
<code>web.fetch</code>	<code>web_fetch</code>
<code>openclaw.status</code>	<code>get_clawdbot_status</code>
<code>openclaw.notifications</code>	<code>get_notifications</code>

eyeterm Control (2 Events)

Event	Tool Function
<code>desktop.eyeterm.toggle</code>	<code>eyeterm_toggle</code>
<code>desktop.eyeterm.calibrate</code>	<code>eyeterm_calibrate</code>

eyeterm Subsystem (15 Directories)

Complete eye tracking system under `python/spaces/desktop/eyeterm/`:

Directory	Purpose
<code>vision/</code>	Gaze tracking (<code>gaze.py</code> 21.2KB), wink detection (<code>wink.py</code> 4.8KB), PolynomialMapper (9.2KB), camera, calibrate
<code>action/</code>	Action execution
<code>ai/</code>	AI-based decisions
<code>audio/</code>	Audio integration
<code>claude/</code>	Claude API integration
<code>cursor/</code>	Gaze-to-cursor mapping
<code>editor/</code>	Editor integration
<code>models/</code>	Data models
<code>routing/</code>	Command routing
<code>screen/</code>	Screen state analysis
<code>stream/</code>	Stream processing
<code>ui/</code>	UI rendering
<code>app.py</code>	Main app (22.8KB)
<code>headless.py</code>	Headless mode (38.6KB)

Directory	Purpose
state.py	State management (5.4KB)

Vision Stack in Detail

- **gaze.py** (21.2KB): MediaPipe FaceMesh (478-landmark model), iris-based gaze tracking, head pose from nose/ears
- **wink.py** (4.8KB): Eye aspect ratio (EAR) calculation, left=confirm / right=cancel, 600ms cooldown
- **polynomial_mapper.py** (9.2KB): Quadratic gaze-to-screen coordinate mapping, click-learning calibration, adaptive fusion (iris+head)

Messaging Pipeline

Complete messaging processing under `python/spaces/desktop/messaging/`:

File	Purpose
incoming_handler.py (8.8KB)	Process incoming messages
messaging_pipeline.py (11.9KB)	Pipeline orchestration
relevance_filter.py (7.3KB)	Relevance filtering

Key Components

Component	File	Purpose
Backend Agent	agents/desktop_agent.py	21 Events → 19 Tools
Desktop Tools	tools/desktop_tools.py	Mouse/keyboard automation
Task Tools	tools/task_tools.py	Task node management
Moire Tools	tools/moire_tools.py	MoireTracker vision
Adapted Tools	tools/adapted_desktop_tools.py	Typed wrappers
Automation UI	Automation_ui/ (submodule)	FastAPI backend for vision agents

Current Status

Implemented

- Eye/gaze tracking via MediaPipe fully functional (478-landmark, iris+head)
- PolynomialMapper for gaze-to-screen mapping (quadratic, click-learning)
- Wink detection (EAR-based, left/right distinction)
- Camera integration and calibration
- 21 events with complete tool mapping

- Messaging pipeline with relevance filtering (WhatsApp, Telegram)
- Moire Tracker vision integration
- Desktop automation (mouse, keyboard, app control)
- Task node management
- Automation UI (FastAPI backend)
- eyeterm app with 15 subsystems

Not Implemented

- OCR engine for text extraction from screen captures
- Video agent for continuous action verification
- ML-based workflow learning (eye tracking data as foundation)
- Agent team for screen validation

Roadmap

- Multi-monitor support and complex desktop layouts
- Privacy-preserving facial analysis (on-device processing)
- Advanced workflow automation based on learned user patterns
- Cross-application context sharing between desktop applications

Ecosystem Fit

Desktop.Space is Vibemind's eyes and hands on the user's computer. It provides context to Ideas.Space about current screen content. It recognizes user activity patterns for The Brain.Space. It executes actions that other spaces orchestrate. It validates that deployments from Coding.Space work correctly. The entire Vibemind platform becomes context-aware through Desktop.Space.